

# Kathryn D. Huff

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| CONTACT INFORMATION                  | <i>Professor</i><br><i>Dept. of Nuclear Engineering and Engineering Physics</i><br><i>University of Wisconsin–Madison</i>                                                                                                                                                                                                                                                                                                                                                           | UW e-mail: <a href="mailto:katy.huff@wisc.edu">katy.huff@wisc.edu</a>                                                |
| RESEARCH INTERESTS                   | Advanced nuclear reactors and fuel cycles, multi-physics simulation, energy systems analysis, scientific computation, nuclear energy policy.                                                                                                                                                                                                                                                                                                                                        |                                                                                                                      |
| PHD                                  | <b>University of Wisconsin–Madison, NUCLEAR ENGINEERING</b><br>• An Integrated Used Fuel Disposition and Generic Repository Model for Fuel Cycle Analysis<br>• Advisor: Professor Paul P.H. Wilson                                                                                                                                                                                                                                                                                  | <b>Aug 2008 – Aug 2013</b>                                                                                           |
| BA                                   | <b>University of Chicago, PHYSICS</b><br>• Celestial Gain Calibrations of QUIET Telescope Polarimeters                                                                                                                                                                                                                                                                                                                                                                              | <b>Aug 2004 – Jun 2008</b>                                                                                           |
| RESEARCH AND PROFESSIONAL EXPERIENCE | <b>University of Wisconsin–Madison, Madison, WI</b><br><i>Professor, Nuclear Engineering and Engineering Physics</i><br><i>Department Chair, Nuclear Engineering and Engineering Physics</i>                                                                                                                                                                                                                                                                                        | <b>July 2026 – Present</b><br><b>July 2026 – Present</b>                                                             |
|                                      | <b>University of Illinois at Urbana-Champaign, Urbana, IL</b><br><i>Associate Professor, Nuclear Plasma and Radiological Engineering</i><br><i>Affiliate Faculty, National Center for Supercomputing Applications</i><br><i>Affiliate Faculty, Computational Science and Engineering</i><br><i>Director, Advanced Reactors and Fuel Cycles group.</i>                                                                                                                               | <b>Nov 2021 – Jun 2026</b><br><b>Aug 2016 – Jun 2026</b><br><b>Aug 2018 – Jun 2026</b>                               |
|                                      | <b>Amazon Web Services, Seattle, WA</b><br><i>Amazon Scholar, Data Center Planning and Delivery, Global Energy Team</i><br><i>Part-time private advisory consulting.</i>                                                                                                                                                                                                                                                                                                            | <b>Aug 2024 – Present</b>                                                                                            |
|                                      | <b>Office of Nuclear Energy, Department of Energy, Washington, DC</b><br><i>Assistant Secretary, Nuclear Energy</i><br><i>Senior Advisor to the Secretary, Nuclear Energy</i><br><i>Acting Assistant Secretary, Nuclear Energy</i><br><i>Principal Deputy Assistant Secretary, Nuclear Energy</i><br><i>Presidentially appointed, Senate Confirmed Official leading the Office of Nuclear Energy</i><br><i>On extended Unpaid Leave of Absence from the University of Illinois.</i> | <b>May 2022 – May 2024</b><br><b>Jan 2022 – May 2022</b><br><b>May 2021 – Jan 2022</b><br><b>May 2021 – Jan 2022</b> |
|                                      | <b>University of Illinois at Urbana-Champaign, Urbana, IL</b><br><i>Blue Waters Assistant Professor</i><br><i>Principal Investigator, Advanced Reactors and Fuel Cycles group.</i>                                                                                                                                                                                                                                                                                                  | <b>Aug 2016 – Sep 2021</b>                                                                                           |
|                                      | <b>University of California - Berkeley, NE Dept., Berkeley, CA</b><br><i>Postdoctoral Scholar, Nuclear Science and Security Consortium</i><br><i>Data Science Fellow, Berkeley Institute for Data Science</i><br><i>Developing computational tools and multiphysics models for advanced reactor safety analysis.</i>                                                                                                                                                                | <b>Sep 2013 – Jul 2016</b><br><b>Aug 2014 – Jul 2016</b>                                                             |
|                                      | <b>Argonne National Laboratory, Argonne, IL</b><br><i>Laboratory Graduate Research Appointee, Used Fuel Disposition Campaign</i><br><i>Developed a used fuel disposition and generic repository computational model.</i>                                                                                                                                                                                                                                                            | <b>Jun 2011 – Aug 2013</b>                                                                                           |
|                                      | <b>University of Wisconsin - Madison, NEEP Dept., Madison, WI</b><br><i>Graduate Research Assistant, Computational Nuclear Engineering Research Group</i><br><i>Developed and applied CYCLUS, a nuclear fuel cycle systems analysis tool.</i>                                                                                                                                                                                                                                       | <b>Jun 2008 – Aug 2013</b>                                                                                           |
|                                      | <b>Idaho National Laboratory, Idaho Falls, ID</b><br><i>Graduate Research Assistant, Systems Analysis Campaign</i>                                                                                                                                                                                                                                                                                                                                                                  | <b>Jun – Aug 2010</b>                                                                                                |

Developed software functions and requirements for the Fuel Cycle Simulator concept.

**Kavli Institute For Cosmological Physics**, Chicago, IL **Jan 2005 – Jun 2008**  
*Research Assistant, Laboratory for Astrophysics and Space Research*  
Programmed & machined instrumentation. Planned protocol for QUIET polarimeter calibration.

**Universidad de Chile, Physics Dept.**, Santiago, Chile **Jun – Sep 2006**  
*Research Assistant, Chicago-Chile Research Exchange Program*  
Constructed and operated a far-from-equilibrium granular materials experiment.

**Los Alamos Neutron Science Center**, Los Alamos, NM **Jun – Sep 2004**  
*Research Assistant, LANSCE-3* **May – Aug 2003**  
Applied digital filtration algorithms and MCNPX models to experimental data.

HONORS AND  
AWARDS

|                                                                                        |                  |
|----------------------------------------------------------------------------------------|------------------|
| NPPE, Students Award for Excellence in Undergraduate Teaching                          | 2017, 2021, 2025 |
| Fin-Erth Women in Climate Award                                                        | 2024             |
| Presidential Citation, American Nuclear Society                                        | 2024             |
| Distinguished Public Service Award, American Nuclear Society (ANS)                     | 2024             |
| 40 Under 40, Nuclear News                                                              | 2024             |
| Meritorious Service Award, United States Nuclear Industry Council                      | 2024             |
| Secretary's Achievement Award, Uranium Strategy Tiger Team, U.S. Dept. of Energy (DOE) | 2024             |
| Secretary's Achievement Award, Pathways to Commercial Liftoff Team, U.S. DOE           | 2023             |
| Warren K. Sinclair Medal, National Council on Radiation Protection & Measurements      | 2023             |
| Presidential Nomination & Senate Confirmation, Assistant Secretary for Nuclear Energy  | 2022             |
| Stanley H. Pierce Award, UIUC Engineering Council                                      | 2019             |
| American Nuclear Society, Oestmann Professional Women's Achievement Award              | 2017             |
| AE3, Collins Scholars Program Graduate                                                 | 2017             |
| UIUC, Teachers Ranked as Excellent                                                     | F 2016, S 2020   |
| American Nuclear Society, Young Member Excellence Award                                | 2016             |
| National Energy Research Scientific Computing Allocation, Senior Investigator          | 2015–2016        |
| Data Science Fellowship, Berkeley Institute for Data Science, UC Berkeley              | 2014–2016        |
| Nuclear Science and Security Consortium Postdoctoral Fellowship, UC Berkeley           | 2013–2016        |
| DOE Office of Science Laboratory Graduate Appointment, Argonne National Lab            | 2011–2013        |
| Roy G Post Foundation Nuclear Waste Management Graduate Scholarship                    | 2011             |
| John Randall Memorial Scholarship, American Nuclear Society FCWMD                      | 2009             |
| J.A McDeavitt Scholarship, University of Chicago, Chicago, IL                          | 2007–2008        |
| University Scholar Award, University of Chicago, Chicago, IL                           | 2004–2008        |
| Los Alamos Distinguished Student Performance Award, Los Alamos National Lab            | 2004             |

GRANTS  
AWARDED

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|--------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Nuclear Science and Security Consortium 4</b>                                                                         | Period: 2026–2031           |
| Source: DOE-NNSA Office of DNN R&D                                                                                       | Award Total: \$25,000,000   |
| Role: Consortium Associate Director, UIUC PI, Thrust Area Lead                                                           | Huff Allocation: ~\$625,000 |
| <b>EINSTEIN: Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides</b>               | Period: 2025–2028           |
| Source: DOE-ARPA-E NEWTON Program                                                                                        | Award Total: \$1,025,405    |
| Role: Collaborator                                                                                                       | Huff Allocation: ~\$250,000 |
| <b>Nuclear Science and Security Consortium 3<sup>1</sup></b>                                                             | Period: 2021–2026           |
| Source: DOE-NNSA Office of DNN R&D                                                                                       | Award Total: \$25,000,000   |
| Role: Consortium Co-PI, UIUC PI, Thrust Area Lead                                                                        | Huff Allocation: \$625,000  |
| <b>Evaluation of micro-reactor requirements and performance in an existing well-characterized micro-grid<sup>1</sup></b> | Period: 2020–2022           |
| Source: DOE-NEUP                                                                                                         | Award Total: \$800,000      |
| Role: Co-PI                                                                                                              | Huff Allocation: \$265,000  |

<sup>1</sup>PI-ship transferred to other leadership in May 2021 corresponding with unpaid leave of absence.

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|------------------------------------------------------------------------------------|-----------------------------------|
| <b>Enabling Load Following Capability in the Transatomic Power MSR<sup>1</sup></b> | Period: 2018–2021                 |
| Source: ARPA - E - MEITNER                                                         | Award Total: <b>\$999,694</b>     |
| Role: Principal Investigator                                                       | Huff Allocation: \$205,000        |
| <b>US Research Software Sustainability Institute (URSSI)</b>                       | Period: 2017–2018                 |
| Source: NSF - OAC - SI2 - S2I2 Conceptualization                                   | Award Total: \$499,999            |
| Role: Senior Personnel                                                             | Huff Allocation: N/A              |
| <b>Dynamic Transition Analysis with TIMES</b>                                      | Period: 2018–2019                 |
| Source: I <sup>2</sup> CNER                                                        | Award Total: \$76,359             |
| Role: Co-PI                                                                        | Huff Allocation: <b>\$76,359</b>  |
| <b>Investigation of Agricultural Uses of Nuclear Waste Heat</b>                    | Period: 2017–2018                 |
| Source: Exelon                                                                     | Award Total: \$151,257            |
| Role: Co-PI                                                                        | Huff Allocation: <b>\$11,678</b>  |
| <b>Consortium for Verification Technology</b>                                      | Period: 2015–2020                 |
| Source: DOE-NNSA Office of DNN R&D                                                 | Award Total: \$25,000,000         |
| Role: Consortium Co-PI, UIUC PI, CVT Investigator                                  | Huff Allocation: <b>\$347,000</b> |
| <b>Consortium for Nonproliferation Enabling Capabilities</b>                       | Period: 2014–2019                 |
| Source: DOE-NNSA Office of DNN R&D                                                 | Award Total: \$25,000,000         |
| Role: Consortium Co-PI, UIUC PI, Thrust Area Lead                                  | Huff Allocation: <b>\$648,000</b> |
| <b>Collaborative, Open-Source Curriculum Development</b>                           | Period: 2017–2018                 |
| Source: UIUC Strategic Instructional Innovations Program                           | Award Total: \$19,347             |
| Role: Principal Investigator                                                       | Huff Allocation: <b>\$13,000</b>  |
| <b>REU Site: INCLUSION at U. Illinois</b>                                          | Period: 2017–2020                 |
| Source: NSF - ACI                                                                  | Award Total: \$380,036            |
| Role: Senior Personnel                                                             | Huff Allocation: N/A              |
| <b>Demand-Driven Cycamore Archetypes</b>                                           | Period: 2016–2019                 |
| Source: DOE, NEUP R&D                                                              | Award Total: \$800,000            |
| Role: Co-PI                                                                        | Huff Allocation: <b>\$395,066</b> |

## BOOKS

- [1] A. M. Scopatz and **K. D. Huff**. *Effective computation in physics: Field guide to research with python*. O'Reilly Media, Sebastopol, CA, 1st edition, May 2015. URL: <http://shop.oreilly.com/product/0636920033424.do>

## BOOK CHAPTERS

- [2] **K. Huff**. Foreword: Global Need for Nuclear Energy New Directions in. In *Nuclear Energy Innovation and Opportunities in Fission and Fusion for Global Decarbonization*, volume 1. 1st edition edition, Jan. 2026. URL: <https://shop.elsevier.com/books/new-directions-in-nuclear-energy/foss/978-0-443-33471-9>
- [3] S. Gesing, M. Pierce, S. Marru, M. Zentner, **K. D. Huff**, S. Bradley, S. B. Cleveland, S. R. Brandt, R. Ramnath, K. Kee, M. Dahan, B. M. V. Martínez, W. C. Sepulveda, and J. J. S. Mondragón. Science Gateways and AI/ML: How Can Gateway Concepts and Solutions Meet the Needs in Data Science? In *Critical Infrastructure - Modern Approach and New Developments*. IntechOpen, Mar. 2023. URL: <https://www.intechopen.com/chapters/86501>, doi:10.5772/intechopen.110144
- [4] **K. Huff**. Chapter One - Economics of Advanced Reactors and Fuel Cycles. In H. Bindra, editor, *Storage and Hybridization of Nuclear Energy*, volume 1, pages 1–20. Science & Technology Books Elsevier, Inc., Cambridge, MA, United States, 1 edition, Jan. 2019. URL: <http://www.sciencedirect.com/science/article/pii/B9780128139752000016>, doi:10.1016/B978-0-12-813975-2.00001-6
- [5] **K. Huff**. Case Study: Cyclus Project. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017
- [6] **K. Huff**. Lessons Learned. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017

- [7] S. G. Dotson, M. Munk, and **K. D. Huff**. Demonstrating the Osier framework for energy system and nuclear fuel cycle optimization. *Annals of Nuclear Energy*, 230:112151, June 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0306454926000393>, doi:10.1016/j.anucene.2026.112151
- [8] S. M. Park, **K. D. Huff**, and M. Munk. Hybrid SN-Diffusion Neutronics Method for Control Rod Modeling in Molten Salt Reactors. *Nuclear Science and Engineering*, 0(0):1–26, May 2026. doi:10.1080/00295639.2026.2667091
- [9] A. M. Bachmann, Z. Richter, **K. D. Huff**, and M. Munk. Effects of Uranium Impurities in Down-blended HEU on HTGR Performance. *Nuclear Technology*, 0(0):1–19, Jan. 2026. doi:10.1080/00295450.2025.2560690
- [10] L. Seifert, S. Shahbazi, S. Richards, M. Munk, and **K. Huff**. Accuracy of the Scaled Flux Method for Spatially Resolved Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors. *Nuclear Science and Engineering*, 0(0):1–13, Sept. 2025. doi:10.1080/00295639.2025.2568258
- [11] L. Seifert, B. Betzler, W. Wieselquist, M. Jessee, M. Munk, and **K. Huff**. Delayed Neutron Precursor Group Parameter and Spectra Generation from Fast Fission of <sup>235</sup>U in SCALE. *Nuclear Science and Engineering*, 0(0):1–16, Aug. 2025. doi:10.1080/00295639.2025.2525754
- [12] W. E. Kennedy, R. A. Meserve, K. A. Higley, **K. D. Huff**, C. T. Hanson, M. Ford, D. Schultheisz, T. Smith, B. Kugelmass, A. Abou-Jaoude, J. R. Lovering, J. D. Semancik, G. V. Cullen, J. Cheatham, P. F. Peterson, E. Redmond II, S. M. Mirsky, M. Mahowald, M. G. Houts, D. Perkins, R. Vaghetto, J. Duhig, and C. J. D. VanHorne-Sealy. National Council on Radiation Protection (NCRP) 2024 annual meeting:advanced and small modular nuclear power reactors. *Journal of Radiological Protection: Official Journal of the Society for Radiological Protection*, 44(4), Oct. 2024. doi:10.1088/1361-6498/ad7ec4
- [13] N. Thiollière, X. Doligez, M. Halasz, G. Krivtchik, I. Merino, B. Mouginot, A. V. Skarbeli, A. Hernandez-Solis, F. Alvarez-Velarde, F. Courtin, H. Druenne, M. Ernoult, **K. Huff**, M. Szieberth, B. Vermeeren, and P. Wilson. Impact of fresh fuel loading management in fuel cycle simulators: A functionality isolation test. *Nuclear Engineering and Design*, 392:111748, June 2022. URL: <https://www.sciencedirect.com/science/article/pii/S0029549322001029>, doi:10.1016/j.nucengdes.2022.111748
- [14] M. Turkmen, G. J. Y. Chee, and **K. D. Huff**. Machine learning application to single channel design of molten salt reactor. *Annals of Nuclear Energy*, 161:108409, Oct. 2021. doi:<https://doi.org/10.1016/j.anucene.2021.108409>
- [15] A. Chaube, A. Chapman, A. Minami, J. Stubbins, and **K. D. Huff**. The role of current and emerging technologies in meeting Japan’s mid- to long-term carbon reduction goals. *Applied Energy*, 304:117669, Dec. 2021. URL: <https://www.sciencedirect.com/science/article/pii/S0306261921010308>, doi:10.1016/j.apenergy.2021.117669
- [16] A. Chapman, Y. Shigetomi, S. Chandra Karmaker, B. Baran Saha, **K. Huff**, C. Brooks, and J. Stubbins. The cultural dynamics of energy: The impact of lived experience, preference and demographics on future energy policy in the United States. *Energy Research & Social Science*, 80:102231, Oct. 2021. URL: <https://www.sciencedirect.com/science/article/pii/S2214629621003248>, doi:10.1016/j.erss.2021.102231
- [17] O. Ashraf, A. Rykhlevskii, G. V. Tikhomirov, and **K. D. Huff**. Preliminary design of control rods in the single-fluid double-zone thorium molten salt reactor (SD-TMSR). *Annals of Nuclear Energy*, 152:108035, Mar. 2021. URL: <http://www.sciencedirect.com/science/article/pii/S0306454920307313>, doi:10.1016/j.anucene.2020.108035
- [18] O. Ashraf, A. Rykhlevskii, G. V. Tikhomirov, and **K. D. Huff**. Strategies for thorium fuel cycle transition in the SD-TMSR. *Annals of Nuclear Energy*, 148, Dec. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0306454920303546>, doi:10.1016/j.anucene.2020.107656
- [19] E. A. Miernicki, A. L. Heald, **K. D. Huff**, C. S. Brooks, and A. J. Margenot. Nuclear waste heat use in agriculture: History and opportunities in the United States. *Journal of Cleaner Production*, 267:121918, Sept. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S095965262031965X>, doi:10.1016/j.jclepro.2020.121918

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- [21] A. Chaube, A. Chapman, Y. Shigetomi, **K. Huff**, and J. Stubbins. The Role of Hydrogen in Achieving Long Term Japanese Energy System Goals. *Energies*, 13(17):4539, Jan. 2020. Number: 17. URL: <https://www.mdpi.com/1996-1073/13/17/4539>, doi:10.3390/en13174539
- [22] J. W. Bae, A. Rykhlevskii, G. J. Y. Chee, and **K. D. Huff**. Deep learning approach to nuclear fuel transmutation in a fuel cycle simulator. *Annals of Nuclear Energy*, 139:107230, May 2020. [github.com/jbae11/depletion\\_rom](https://github.com/jbae11/depletion_rom). URL: <http://www.sciencedirect.com/science/article/pii/S0306454919307406>, doi:10.1016/j.anucene.2019.107230
- [23] O. Ashraf, A. Rykhlevskii, G. Tikhomirov, and **K. D. Huff**. Whole core analysis of the single-fluid double-zone thorium molten salt reactor (SD-TMSR). *Annals of Nuclear Energy*, 137:107–115, Mar. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919306255>, doi: <https://doi.org/10.1016/j.anucene.2019.107115>
- [24] M. Kamuda, J. Zhao, and **K. Huff**. A comparison of machine learning methods for automated gamma-ray spectroscopy. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 954:161385, Feb. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0168900218313779>, doi:10.1016/j.nima.2018.10.063
- [25] J. W. Bae, C. E. Singer, and **K. D. Huff**. Synergistic spent nuclear fuel dynamics within the European Union. *Progress in Nuclear Energy*, 114:1–12, July 2019. URL: <https://www.sciencedirect.com/science/article/pii/S014919701930037X>, doi:10.1016/j.pnucene.2019.02.001
- [26] J. W. Bae, J. L. Peterson-Droogh, and **K. D. Huff**. Standardized verification of the Cyclus fuel cycle simulator. *Annals of Nuclear Energy*, 128:288–291, June 2019. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919300179>, doi:10.1016/j.anucene.2019.01.014
- [27] A. Rykhlevskii, J. W. Bae, and **K. D. Huff**. Modeling and simulation of online reprocessing in the thorium-fueled molten salt breeder reactor. *Annals of Nuclear Energy*, 128:366–379, June 2019. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919300350>, doi:10.1016/j.anucene.2019.01.030
- [28] A. Lindsay, G. Ridley, A. Rykhlevskii, and **K. Huff**. Introduction to Moltres: An application for simulation of Molten Salt Reactors. *Annals of Nuclear Energy*, 114:530–540, Apr. 2018. URL: <https://www.sciencedirect.com/science/article/pii/S0306454917304760>, doi:10.1016/j.anucene.2017.12.025
- [29] A. M. Smith, K. E. Niemeyer, D. S. Katz, L. A. Barba, G. Githinji, M. Gymrek, **K. D. Huff**, C. R. Madan, A. C. Mayes, K. M. Moerman, P. Prins, K. Ram, A. Rokem, T. K. Teal, R. V. Guimera, and J. T. Vanderplas. Journal of Open Source Software (JOSS): design and first-year review. *PeerJ Computer Science*, 4:e147, Feb. 2018. URL: <https://peerj.com/articles/cs-147>, doi:10.7717/peerj-cs.147
- [30] A. Lindsay and **K. Huff**. Moltres: finite element based simulation of molten salt reactors. *The Journal of Open Source Software*, 3(21):1–2, Jan. 2018. doi:10.21105/joss.00298
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- [33] C. Andreades, A. T. Cisneros, J. K. Choi, A. Y. Chong, M. Fratoni, S. Hong, L. R. Huddar, **K. D. Huff**, J. Kendrick, D. L. Krumwiede, M. Laufer, M. Munk, R. O. Scarlat, X. Wang, N. Zwiebaum, E. Greenspan, and P. Peterson. Design Summary of the Mark-I Pebble-Bed, Fluoride Salt-Cooled,

High-Temperature Reactor Commercial Power Plant. *Nuclear Technology*, 195(3):222–238, Sept. 2016. URL: <http://www.ans.org/pubs/journals/nt/a.38935>, doi:10.13182/NT16-2

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- REFEREED  
CONFERENCE  
PROCEEDINGS [37] M. Fang, **K. Huff**, T. Kozlowski, J. F. Stubbins, L. Vergari, and A. D. Fulvio. EINSTEIN: enhanced integrated nuclear systems for transmutation and efficient isolation of nuclides. In *Hard X-Ray, Gamma-Ray, and Neutron Detector Physics XXVII*, volume 13621, pages 128–133. SPIE, Sept. 2025. doi:10.1117/12.3067982
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- [210] **K. D. Huff.** Prospects for Advanced Nuclear Energy - Closing Plenary. In *NEST Small Modular Reactor Prize Competition*, Aug. 2021. The Nuclear Education, Skills and Technology (NEST) Framework SMR Project is an ongoing collaboration between academic and research institutions in Belgium, Canada, France, Switzerland, and the United States. URL: <https://smrprize.com/>
- [211] **K. D. Huff.** Diversity, Equity, and Inclusion Efforts at the U.S. Department of Energy’s Office of Nuclear Energy. In *Modeling, Experimentation and Validation (MeV) Summer School*, Argonne National Laboratory, July 2021. URL: <https://www.anl.gov/article/argonne-hosts-summer-school-for-nuclear-energy-scientists>

- [212] **K. D. Huff.** Mentoring the next generation of scientists in Kenya - Opening Remarks. In *OECD-NEA, IFNEC, NuPEA International Mentoring Workshop in Science, Technology, Engineering and Mathematics*, Kenya, July 2021. URL: [https://www.oecd-nea.org/jcms/pl\\_60344/mentoring-the-next-generation-of-scientists-in-kenya](https://www.oecd-nea.org/jcms/pl_60344/mentoring-the-next-generation-of-scientists-in-kenya)
- [213] **K. D. Huff.** Department of Energy Priorities - Keynote. In *Nuclear Energy Institute Fuel Supply Forum*, Virtual, July 2021
- [214] **K. D. Huff.** EnergyInnovates: Challenges and Opportunities Facing Nuclear Power - Panel. In *U.S. Chamber of Commerce EnergyInnovates Session*, Virtual, July 2021. URL: <https://www.youtube.com/watch?v=IG202YjiAeI>
- [215] **K. D. Huff.** Nuclear Energy Policy. In *U.S. DOE Policy Bootcamp*, United States Department of Energy, Aug. 2021
- [216] **K. D. Huff.** Small Modular Reactors and Finance - Keynote. In *Eurasia Green Energy and Clean Technologies Forum Day 2: Small Modular Reactors and Finance*, Virtual, July 2021. U.S. Department of Commerce, International Trade Administration, and U.S. Commercial Service. URL: <https://www.solar.ist/wp-content/uploads/2021/06/EurasiaGreenEnForumFlyer.pdf>
- [217] **K. D. Huff.** Accident Tolerant Fuel R&D at the Department of Energy. In *Nuclear Energy Institute Accident Tolerant Fuel Working Group*, Virtual, July 2021
- [218] **K. D. Huff.** Advanced Reactor R&D in the Department of Energy - Remarks with Q&A session. In *Nuclear Energy Institute Advanced Reactor Forum*, Virtual, June 2021. URL: <https://www.gotomeet.me/EverettRedmond/nei-arforum>
- [219] **K. D. Huff.** Multiphysics in Fluid-Fueled Reactors - Seminar. In *Oregon State University Nuclear Science and Engineering Seminar*, Corvallis, OR, May 2021. URL: <https://oregonstate.zoom.us/j/98589105007?pwd=MThpb1htM0xhR1lr2c4a1NPK0lvZz09>
- [220] **K. Huff.** So You Want to Get Paid to Save the World? - Panel. In *The 2021 ARPA-E Energy Innovation Summit Student Program*, Virtual, May 2021. URL: <https://sites.duke.edu/gradstudentopps/files/2021/04/Student-Summit-Programming-One-Pager.pdf>
- [221] **K. D. Huff.** Multiphysics in Fluid-Fueled Reactors - Seminar. In *Oregon State University Nuclear Science and Engineering Seminar*, Corvallis, OR, May 2021. URL: <https://oregonstate.zoom.us/j/98589105007?pwd=MThpb1htM0xhR1lr2c4a1NPK0lvZz09>
- [222] **K. D. Huff.** Nuclear Energy Opportunities at DOE Sites. In *2021 National Cleanup Workshop: Capitalizing on a New Era of Cleanup Success*, Alexandria, VA, Dec. 2021. URL: <https://www.cleanupworkshop.com/2021-proceedings.html>
- [223] **K. D. Huff.** The Essential Role of Universities in Advanced Reactor Deployment. In *Spring Nuclear Engineering Colloquium Series*, University of California - Berkeley, Berkeley, CA, Jan. 2021. URL: <https://nuc.berkeley.edu/the-essential-role-of-universities-in-advanced-reactor-deployment/>
- [224] **K. D. Huff.** Customer Perspective – Panel. In *Invited Panelist, GAIN-EPRI-NEI*, Virtual, Aug. 2020
- [225] **K. D. Huff.** Effective Writing (at home?). In *Society of Women Engineers*, Virtual, May 2020. URL: <https://katyhuff.github.io/2020-05-20-writing/>
- [226] **K. D. Huff.** SIAM CSE: Computational Methods for Kinetic Models of Gases and Plasmas Minisymposium - Invited Speaker. In *SIAM CSE 2019*, Feb. 2019
- [227] **K. D. Huff.** Conference Keynote. In *Hack Illinois*, University of Illinois at Urbana-Champaign, Urbana, IL, Feb. 2018
- [228] **K. D. Huff.** Modeling & Simulation of Advanced Reactors and Fuel Cycles. In *Nuclear Engineering and Radiological Sciences*, University of Michigan, Ann Arbor, MI, United States, Feb. 2018. URL: <https://events.umich.edu/event/48405>
- [229] **K. D. Huff.** Doing Our Best: Practices in Scientific Computing. In *PyData Ann Arbor*, Ann Arbor, MI, Feb. 2018. URL: <https://www.youtube.com/watch?v=wkboQuY8Tjg>

- [230] **K. D. Huff.** Why Nuclear? Invited Lecture. In *Olin College of Engineering*, Olin College, Neeham, MA, Oct. 2017
- [231] **K. D. Huff.** Seminar. In *NNSA Nuclear Nonproliferation*, Argonne National Laboratory, Lemont, IL, Sept. 2017
- [232] **K. D. Huff.** Keynote. In *Scientific Python Conference*, Austin, TX, United States, July 2017. URL: <https://scipy2017.scipy.org/>
- [233] **K. D. Huff.** Workforce Transition: How to Succeed-Panel. In *Panel, Young Members Group*, San Francisco, CA, June 2017. URL: <https://epubs.ans.org/?a=40958>
- [234] **K. D. Huff**, J. W. Bae, K. A. Mummah, R. R. Flanagan, and A. M. Scopatz. Current Status of Predictive Transition Capability in Fuel Cycle Simulation. In *Proceedings of Global 2017*, Seoul, South Korea, Sept. 2017
- [235] **K. D. Huff.** Keynote. In *Scientific Python Conference*, Austin, TX, United States, July 2017. URL: <https://scipy2017.scipy.org/>
- [236] **K. D. Huff.** Keynote. In *PyData 2016, Chicago, IL*, Chicago, IL, USA, Aug. 2016. URL: <https://pyvideo.org/pydata-2016/keynote-katy-huff.html>
- [237] **K. D. Huff.** A Computational Future for Science Education. In *Computational, Mathematics, Science & Engineering Seminar, Michigan State University*, East Lansing, MI, USA, Dec. 2015. URL: <https://katyhuff.github.io/2015-11-06-ncsa/>
- [238] **K. D. Huff.** Seminar. In *NNSA Nuclear Nonproliferation*, Argonne National Laboratory, Lemont, IL, Sept. 2017
- [239] **K. D. Huff.** Seminar. In *RPNSD Seminar*, Oak Ridge National Laboratory, Oak Ridge, TN, USA, June 2017
- [240] **K. D. Huff.** Seminar. In *Mechanical and Aerospace Engineering Seminar*, UC Davis, Davis, CA, USA, Apr. 2017
- [241] **K. D. Huff.** Lightning Talk. In *AE3 Lightning Symposium*, University of Illinois, Urbana-Champaign, IL, USA, Mar. 2017
- [242] **K. D. Huff.** Undergraduate Seminar. In *Nuclear, Plasma, & Radiological Engineering Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Feb. 2017
- [243] **K. D. Huff.** Symposium. In *Berkeley Institute for Data Science Symposium*, UC Berkeley, Berkeley, CA, USA, Jan. 2017
- [244] **K. D. Huff.** Seminar. In *Informatics Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Oct. 2016
- [245] **K. D. Huff.** Seminar. In *RPNSD Seminar*, Oak Ridge National Laboratory, Oak Ridge, TN, USA, Mar. 2016
- [246] **K. D. Huff.** Seminar. In *Nuclear Engineering Seminar*, University of Tennessee, Knoxville, Knoxville, TN, USA, Mar. 2016
- [247] **K. D. Huff.** Seminar. In *Nuclear, Plasma & Radiological Engineering Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Dec. 2015
- [248] **K. D. Huff.** Keynote. In *Python in High Performance Computing Workshop, SC15*, Austin, TX, USA, Nov. 2015
- [249] **K. D. Huff.** Colloquium. In *NCSA Colloquium*, University of Illinois, Urbana-Champaign, IL, USA, Nov. 2015
- [250] **K. D. Huff.** Colloquium. In *Nuclear Engineering Colloquium*, North Carolina State University, Raleigh, NC, USA, Oct. 2015
- [251] **K. D. Huff.** Colloquium. In *Nuclear Engineering Colloquium*, Texas A&M University, College Station, TX, USA, Sept. 2015

- [252] **K. D. Huff.** Invited Colloquium. In *Mechanical & Nuclear Engineering Colloquium, Rensselaer Polytechnic Institute*, Troy, NY, USA, Sept. 2015
- [253] **K. D. Huff.** What Can Academia Learn from Open Source? - Panel. In *What Can Academia Learn from Open Source?*, University of Washington, Seattle, WA, USA, Feb. 2015. URL: <https://ti.to/github-events/academia-town-hall->

ENGINEERING  
TEACHING

**University of Illinois at Urbana-Champaign**

DEPT. OF NUCLEAR, PLASMA, AND RADIOLOGICAL ENGINEERING

NPRE 100, *Introduction to Nuclear, Plasma, and Radiological Engineering*

Fall 2024

Fall 2025

NPRE 247, *Modeling Nuclear Energy Systems*

Fall 2018

NPRE 412, *Nuclear Power Economics and Fuel Management*

Fall 2016

Fall 2017

Spring 2020

Spring 2021

Spring 2025

Spring 2026

NPRE 446, *Radiation Interactions with Matter I*

Fall 2019

NPRE 555, *Reactor Theory I*

Spring 2018

Fall 2020

Fall 2025

NPRE 560, *Reactor Kinetics and Dynamics*

Spring 2019

Spring 2021

Fall 2024

GUEST  
LECTURES

**University of California, Berkeley**, DEPT. OF NUCLEAR ENGINEERING

Nov 10, 2020

NE 100, *Introduction to Nuclear Engineering*

Nuclear Fuel Cycle, Advanced Reactors

**University of California, Berkeley**, DEPT. OF NUCLEAR ENGINEERING

Apr 1,3,22, 2015

NE 155, *Introduction to Numerical Simulations in Radiation Transport* Point Reactor Kinetics, Monte Carlo Methods

**University of California, Berkeley**, DEPT. OF NUCLEAR ENGINEERING

Sep 11, 2014

NE 255, *Numerical Simulation in Radiation Transport*

Best Practices in Computational Nuclear Engineering

**University of Wisconsin - Madison**, DEPT. OF NUCLEAR ENGINEERING

Apr 1&3, 2013

NE 571, *Economic and Environmental Aspects of Nuclear Energy*

Nuclear Waste Repository Technology, Policy, and History

**University of Wisconsin - Madison**, DEPT. OF NUCLEAR ENGINEERING

Sep 9&11, 2009

NE 406, *Nuclear Reactor Analysis*

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

**University of Wisconsin - Madison**, DEPT. OF NUCLEAR ENGINEERING

Feb 10, 2010

NE 506, *Practicum in Monte Carlo Radiation Transport*

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

INVITED  
SCIENTIFIC  
COMPUTING  
TEACHING

**SciPy Conference**, Austin, TX

Jul 6–7, 2015

# Introductory Python For Scientific Software

**University of Split**, Split, Croatia **Sep 8–13, 2014**  
G-Node Advanced Scientific Programming in Python Summer School

**SciPy Conference**, Austin, TX **Jun 25, 2013**  
Version Control and Unit Testing For Scientific Software

**University of Chicago, Graduate School**, Chicago, IL **Jan 12–13, 2013**  
Computational Literacy Workshop

**University of California, Berkeley**, Berkeley, CA **Oct 20–21, 2012**  
Department of Statistics Scientific Computing Workshop

**Lawrence Berkeley National Laboratory**, Berkeley, CA **Oct 17–18, 2012**  
Software Carpentry Python Workshop

**International Center for Theoretical Physics**, Trieste, Italy **Feb 20–Mar 2, 2012**  
UNESCO/IAEA Advanced School on Scientific Software Development

**University of Toronto**, Toronto, ON, Canada **Nov 7–8, 2011**  
SciNet Consortium For High Performance Computing Software Carpentry Bootcamp

**American Nuclear Society Winter Meeting**, Washington, D.C. **Nov 1, 2011**  
Young Professionals Congress Hacker Within Scientific Computing Tutorial

**Michigan State University**, East Lansing, MI **Jun 4–5, 2011**  
Institute for Cyber Enabled Research (iCER) and BEACON Center THW Bootcamp

## SCIENTIFIC COMPUTING TEACHING

**Berkeley Institute for Data Science**, Berkeley, CA **Jan 14–15, 2015**  
Managing Databases in SQL

**Berkeley Institute for Data Science**, Berkeley, CA **Jun 4–5, 2015**  
Testing for Scientific Software

**Lawrence Berkeley National Laboratory**, Berkeley, CA **Apr 14–15, 2014**  
Women in Science and Engineering Bootcamp

**The University of Chicago**, Chicago, IL **Apr 2–3, 2012**  
Software Carpentry Scientific Computing Workshop

**The University of Wisconsin**, Madison, WI **Jan 12–14, 2011**  
The Hacker Within Software Carpentry Bootcamp

**The University of Wisconsin**, Madison, WI **Jan 12–14, 2010**  
The Hacker Within Python Bootcamp

**The University of Wisconsin**, Madison, WI **Mar 24–31, 2009**  
The Hacker Within C++ Bootcamp

**The University of Wisconsin**, Madison, WI **Jan 12–15, 2009**  
University of Wisconsin, Hacker Within UNIX Bootcamp

## POSTDOCTORAL RESEARCHERS

| <u>NAME</u>       | <u>DATES</u> | <u>ROLE</u> |
|-------------------|--------------|-------------|
| Mehmet Turkmen    | 2019–2020    | Advisor     |
| Alexander Lindsay | 2016–2017    | Advisor     |

## GRADUATE RESEARCHERS

| <u>NAME</u>        | <u>DEGREE - YEAR</u> | <u>ROLE</u>      |
|--------------------|----------------------|------------------|
| Michael Cheng      | MS - 2017            | MS Second Reader |
| Mark Kamuda        | MS - 2017            | MS Second Reader |
| Mark Kamuda        | PhD - 2019           | PhD Advisor      |
| Gregory Westphal   | MS - 2019            | MS Advisor       |
| Jin Whan Bae       | MS - 2019            | MS Advisor       |
| Erik Medhurst      | MS - 2020            | MS Advisor       |
| Andrei Rykhlevskii | PhD - 2020           | PhD Advisor      |

|                                 |                   |                              |
|---------------------------------|-------------------|------------------------------|
| <b>Katherine C. Hepler</b>      | PhD - 2020        | Dissertation Committee Chair |
| <b>Alvin Lee</b>                | MS - 2020         | MS Second Reader             |
| <b>Roberto Fairhurst-Agosta</b> | MS 2020           | PhD Advisor                  |
| <b>Anshuman Chaube</b>          | MS - 2021         | PhD Advisor                  |
| <b>Gwendolyn Chee</b>           | PhD - 2022        | PhD Co-Advisor               |
| <b>Amanda Bachmann</b>          | PhD - 2023        | PhD Co-Advisor               |
| <b>Sun Myung Park</b>           | PhD - 2024        | PhD Advisor                  |
| <b>Nathan Ryan</b>              | MS - 2024         | MS Advisor                   |
| <b>Samuel Dotson</b>            | PhD - 205         | PhD Advisor                  |
| <b>Zoë Richter</b>              | PhD - (est. 2026) | PhD Advisor                  |
| <b>Luke Seifert</b>             | PhD - (est. 2026) | PhD Advisor                  |
| <b>Oleksandr Yardas</b>         | PhD - (est. 2026) | PhD Advisor                  |
| <b>Elijah Capps</b>             | PhD - (est. 2029) | PhD Advisor                  |
| <b>Liam Pohlmann</b>            | PhD - (est. 2029) | PhD Advisor                  |
| <b>Samar Abdelkawy</b>          | MS - (est. 2026)  | MS Advisor                   |

| UNDERGRADUATE<br>RESEARCHERS | <u>NAME</u>              | <u>DEGREE - YEAR</u> | <u>SCHOLARSHIPS</u>                                                             |
|------------------------------|--------------------------|----------------------|---------------------------------------------------------------------------------|
|                              | <b>Jin Whan Bae</b>      | BS - 2017            | NPPE Outstanding Undergrad Research<br>ANS Best Student Fuel Cycle Presentation |
|                              | <b>Kathryn Mummah</b>    | BS - 2017            | Roy G. Post Foundation Scholarship<br>ANS FCWMD Randall Scholar                 |
|                              | <b>Eric Riewski</b>      | BS - 2017            |                                                                                 |
|                              | <b>GyuTae Park</b>       | BS - (est. 2018)     |                                                                                 |
|                              | <b>Yukun Tan</b>         | BS - (est. 2018)     | Students Pushing Innovation                                                     |
|                              | <b>Lu Kissinger</b>      | BS - 2019            |                                                                                 |
|                              | <b>Xin Wen</b>           | BS - 2018            | Students Pushing Innovation                                                     |
|                              | <b>Daniel Chiu</b>       | BS - 2020            |                                                                                 |
|                              | <b>Tyler Kennelly</b>    | BS - 2019            |                                                                                 |
|                              | <b>Bradley Ellis</b>     | BS - 2019            |                                                                                 |
|                              | <b>Adam Pichman</b>      | BS - 2019            |                                                                                 |
|                              | <b>Zoë Richter</b>       | BS - 2018            |                                                                                 |
|                              | <b>Gavin Davis</b>       | BS - 2021            |                                                                                 |
|                              | <b>Kip Kleimenhagen</b>  | BS - 2021            |                                                                                 |
|                              | <b>David Atwater</b>     | BS - 2021            |                                                                                 |
|                              | <b>Nathan Ryan</b>       | BS - 2022            |                                                                                 |
|                              | <b>Anna Balla</b>        | BS - 2021            |                                                                                 |
|                              | <b>Nataly Panczyk</b>    | BS - 2024            |                                                                                 |
|                              | <b>Bella Pequette</b>    | BS - 2025            |                                                                                 |
|                              | <b>Cesar Zambrano</b>    | BS - 2025            |                                                                                 |
|                              | <b>Rhys MacMillan</b>    | BS - (est. 2026)     |                                                                                 |
|                              | <b>Owen Strong</b>       | BS - (est. 2028)     |                                                                                 |
|                              | <b>Matthew Disimone</b>  | BS - (est. 2027)     |                                                                                 |
|                              | <b>Bryan Park</b>        | BS - (est. 2027)     |                                                                                 |
|                              | <b>Thomas Bohnert</b>    | BS - (est. 2028)     |                                                                                 |
|                              | <b>Madeleine Wallman</b> | BS - (est. 2028)     |                                                                                 |

| VISITING<br>RESEARCHERS | <u>NAME</u>              | <u>DATES</u> | <u>LEVEL - INSTITUTION</u>                    |
|-------------------------|--------------------------|--------------|-----------------------------------------------|
|                         | <b>Gavin Ridey</b>       | 2017         | BS–University of Tennessee, Knoxville         |
|                         | <b>Aditya Bhosale</b>    | 2017         | BS - IIT, Bombay                              |
|                         | <b>Snehal Chandan</b>    | 2017         | BS - IIT, Bombay                              |
|                         | <b>Eleonora Skrzypek</b> | 2019         | PhD - Warsaw University of Technology, Poland |

|                                   |                      |                                           |
|-----------------------------------|----------------------|-------------------------------------------|
| SCIENTIFIC<br>COMPUTING<br>SKILLS | <b>Languages</b>     | bash/csh, C++, FORTRAN, Perl, Python, XML |
|                                   | <b>Build Systems</b> | make, CMake, automake                     |
|                                   | <b>Databases</b>     | HDF5, SQL                                 |

|                          |                                                                                      |                                                                                             |
|--------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                          | <b>Test Frameworks</b>                                                               | CTest, GoogleTest, nose                                                                     |
|                          | <b>Version Control</b>                                                               | cvs, git, hg, svn                                                                           |
|                          | <b>Other Tools</b>                                                                   | Doxygen, Sphinx, GoldSim, L <sup>A</sup> T <sub>E</sub> X, Mathematica, MatLab, MCNP, MOOSE |
| EDITING AND<br>REVIEWING | <b>Editorial Board</b>                                                               | <i>Nuclear Technology</i> <b>2024 – present</b>                                             |
|                          |                                                                                      | <i>Journal of Open Source Education</i> <b>2018 – 2021</b>                                  |
|                          |                                                                                      | <i>Journal of Open Source Software</i> <b>2016 – 2021</b>                                   |
|                          |                                                                                      | <i>Nuclear Technology</i> <b>2018 – 2021</b>                                                |
|                          |                                                                                      | <i>Nuclear Engineering and Design</i> <b>2020 – 2021</b>                                    |
|                          |                                                                                      | <i>Papers in Physics</i> <b>2020 – 2023</b>                                                 |
|                          |                                                                                      | <i>Proceedings of the SciPy Scientific Python Conference</i> <b>2013, 2015, &amp; 2017</b>  |
|                          | <b>Manuscript Referee</b>                                                            | <i>Annals of Nuclear Energy</i>                                                             |
|                          |                                                                                      | <i>Journal of Nuclear Energy Science and Power Generation Technology</i>                    |
|                          |                                                                                      | <i>Nuclear Engineering and Design</i>                                                       |
|                          |                                                                                      | <i>Nuclear Science and Engineering</i>                                                      |
|                          |                                                                                      | <i>Nuclear Technology</i>                                                                   |
|                          |                                                                                      | <i>Progress in Nuclear Energy</i>                                                           |
|                          | <b>Grant Proposal Referee</b>                                                        | <i>Dept. of Energy Nuclear Energy University Programs</i>                                   |
|                          |                                                                                      | <i>Dept. of Energy Technology Commercialization Fund</i>                                    |
|                          |                                                                                      | <i>Blue Waters Fellows Program</i>                                                          |
|                          |                                                                                      | <i>Alfred P. Sloan Foundation</i>                                                           |
|                          | <b>Book Proposal Referee</b>                                                         | <i>O'Reilly Media</i>                                                                       |
|                          |                                                                                      | <i>Elsevier</i>                                                                             |
| PROFESSIONAL<br>SERVICE  | <b>Member</b> , Argonne Board of Governors Program Review for Nuclear Energy         | <b>2025–present</b>                                                                         |
|                          | <b>Member</b> , Advocacy Council, Nuclear Matters, Washington D.C.                   | <b>2025–present</b>                                                                         |
|                          | <b>Member</b> , External Affairs Committee, ANS                                      | <b>2025–present</b>                                                                         |
|                          | <b>Vice Chair</b> , Mathematics and Computation Division, ANS                        | <b>2025–present</b>                                                                         |
|                          | <b>Advisory Committee</b> , Digital Information Technology, Sloan Foundation         | <b>2019–2021</b>                                                                            |
|                          | <b>Chair</b> , Nonproliferation and Policy Division, ANS                             | <b>2020–2021</b>                                                                            |
|                          | <b>Executive Committee</b> , Mathematics and Computation Division, ANS               | <b>2020–2021</b>                                                                            |
|                          | <b>Vice Chair</b> , Nonproliferation and Policy Division, ANS                        | <b>2019–2020</b>                                                                            |
|                          | <b>Chair &amp; Host</b> , Technical Workshop on Fuel Cycle Simulation                | <b>2019</b>                                                                                 |
|                          | <b>Past Chair</b> ( <i>ex officio</i> ), Fuel Cycle & Waste Management Division, ANS | <b>2016–2017</b>                                                                            |
|                          | <b>Co-Organizer</b> , Technical Workshop on Fuel Cycle Simulation                    | <b>2017</b>                                                                                 |
|                          | <b>Technical Program Committee</b> , IHLRWM Conference                               | <b>2017</b>                                                                                 |
|                          | <b>Chair</b> , Fuel Cycle & Waste Management Division, ANS                           | <b>2016–2017</b>                                                                            |
|                          | <b>Vice Chair</b> , Fuel Cycle & Waste Management Division, ANS                      | <b>2015–2016</b>                                                                            |
|                          | <b>Chair</b> , Steering Committee, Software Carpentry Foundation                     | <b>2014–2015</b>                                                                            |
|                          | <b>Secretary–Treasurer</b> , Fuel Cycle & Waste Management Division, ANS             | <b>2013–2015</b>                                                                            |
|                          | <b>Secretary</b> , Young Members Group, ANS                                          | <b>2013–2014</b>                                                                            |
|                          | <b>Technical Program Co-Chair</b> , SciPy, Scientific Python Conference              | <b>2013–2014</b>                                                                            |
|                          | <b>Member</b> , Next Generation Leadership Committee, Waste Management Symposium     | <b>2013–2014</b>                                                                            |
|                          | <b>Moderator, Organizer, Panelist</b> , inSCiight Scientific Computing Podcast       | <b>2011–2013</b>                                                                            |
|                          | <b>Co-Founder</b> , Nuclear Pride, LGBTQA Organization                               | <b>2011–2013</b>                                                                            |
|                          | <b>Co-Founder, Treasurer, President</b> , Hacker Within Scientific Computing Group   | <b>2008–2011</b>                                                                            |
|                          | <b>Governor, Treasurer</b> , University of Wisconsin ANS student section             | <b>2008–2010</b>                                                                            |
| DEPARTMENTAL<br>SERVICE  | <b>Graduate Committee</b> , Qualifying Exam Sub-Committee                            | <b>2024–present</b>                                                                         |
|                          | <b>Graduate Committee</b> , Admissions Sub-Committee                                 | <b>Spring 2024</b>                                                                          |

|                    |                                                                                                      |                     |
|--------------------|------------------------------------------------------------------------------------------------------|---------------------|
|                    | <b>Faculty Advisor, UIUC ANS Student Section</b>                                                     | <b>2024–present</b> |
|                    | <b>Faculty Advisor, UIUC ANS Student Section</b>                                                     | <b>2016–2021</b>    |
|                    | <b>Undergraduate Committee,</b>                                                                      | <b>2019–2021t</b>   |
|                    | <b>Graduate Committee, Qualifying Exam Sub-Committee</b>                                             | <b>2017-2019</b>    |
|                    | <b>Graduate Committee, Admissions Sub-Committee</b>                                                  | <b>Spring 2017</b>  |
|                    | <b>Graduate Committee, Admissions Sub-Committee</b>                                                  | <b>Fall 2016</b>    |
|                    | <b>Advisory Committee,</b>                                                                           | <b>2017–2018</b>    |
|                    | <b>Faculty Search Committee,</b>                                                                     | <b>2017–2018</b>    |
|                    | <b>Faculty Advisor, UIUC WiN Student Section</b>                                                     | <b>2017–2018</b>    |
| COLLEGE<br>SERVICE | <b>Member, Courses and Curriculum Committee</b>                                                      | <b>2025–present</b> |
|                    | <b>Fellow, Education Innovation Fellows</b>                                                          | <b>2025–present</b> |
|                    | <b>Member, Instructional Facility Working Group</b>                                                  | <b>2017–2018</b>    |
|                    | <b>Selection Committee, Clare Boothe Luce (CBL) Research Scholars</b>                                | <b>2020–2021</b>    |
|                    | <b>Member, Engineering IT Governance Education Working Group</b>                                     | <b>2020–2021</b>    |
|                    | <b>Faculty Mentor, ARISE program</b>                                                                 | <b>2019–2020</b>    |
|                    | <b>Member, ENG/TE Liaison Committee</b>                                                              | <b>2020–present</b> |
|                    | <b>Member, Instructional Facility Working Group</b>                                                  | <b>2017–2018</b>    |
|                    | <b>Faculty Advisor, UIUC CSE The Hacker Within Scientific Computing Group</b>                        | <b>2016–2017</b>    |
| CAMPUS<br>SERVICE  | <b>Senator, U. Illinois at Urbana-Champaign Faculty Senate</b>                                       | <b>2025–present</b> |
|                    | <b>Steering Committee Member, Illinois Data Science Initiative</b>                                   | <b>2018</b>         |
|                    | <b>Hack Mentor, Hack Illinois</b>                                                                    | <b>2017</b>         |
| CONSULTING         | <b>Amazon Web Services</b> Seattle, WA                                                               | <b>2024–Present</b> |
|                    | <i>Amazon Scholar</i><br>Nuclear energy due dilligence, data center planning and delivery            |                     |
|                    | <b>Thomas Edison State University</b> Trenton, NJ                                                    | <b>2018–2019</b>    |
|                    | <i>Subject Matter Expert</i><br>Institute of Nuclear Power Operations (INPO) Academic Program Review |                     |